walk for ever would return to the point on the Earth from which he departed.' In other words, he adopts an internal, not an external, point of view: the perspective of someone who walks the Earth, not of someone who looks at it from afar. At first glance it might seem like a pointless, complicated way of explaining that the Earth is a ball. Why doesn't Brunetto simply say that the Earth is like an orange? But, on reflection: if, say, an ant walks on an orange, it will at some point find itself upside down, and must keep itself attached by means of the tiny suction pads on its legs, to avoid falling off. And yet a traveller who walks the Earth never finds himself upside down, and needs no suction pads on his legs. Brunetto's description is not so quaint after all.

Now, think about it. For someone who has learned from his teacher that the form of the surface of our planet is such that by walking always in a straight line we return to the point we started from, it is perhaps not so difficult to take the next obvious step, and imagine that the form of the entire universe is such that, flying always in a straight line, we return to the same point of departure: a 3-sphere is a space in which 'two winged knights that could *fly* in opposite directions would meet up on the other side'. In technical terms, the description of the geometry of the Earth offered by Brunetto Latini in *Li tresor* is given in terms of intrinsic geometry (seen from the inside) rather than extrinsic (seen from the outside), and this is exactly the description that is suitable to generalize the notion of 'sphere' from two dimensions to three. The best way of describing a 3-sphere is not to try to 'see it from the outside', but rather to describe what happens when moving within it.

The method developed by Gauss to describe curved surfaces, and generalized by Riemann to describe the curvature of spaces in three or more dimensions, basically, amounts to Brunetto Latini's way. That is to say, the idea is to describe a curved space not as 'seen from the outside', stating how it curves in an external space, but instead in terms of what may be experienced by somebody *within* that space, who is moving and always remaining within it. For instance, the surface of an ordinary sphere, as Brunetto observes, is a surface where all the 'straight' lines get back to the starting point after traversing the same distance (the length of the equator). A 3-sphere is a three-dimensional space with the same property.

Einstein's spacetime is not curved in the sense that it curves 'in an external space'. It is curved in the sense that its *intrinsic* geometry, that is to say, the web of distances between its points, which can be observed by staying within it, is not the geometry of a flat space. It is a space where Pythagoras's theorem is not valid, just as Pythagoras's theorem is not valid on the surface of the Earth.[fn19]

There is a way of understanding the curvature of space from within it, and without looking at it from outside, which is important for what follows. Imagine you are at the North Pole and walk southwards until you reach the